Question:

Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think>
the initial cost of planting the tree is 90 dollars
the tree grows 7 lemons per year, and each lemon sells for 1.5 dollars, so the total income from lemons per year is 7 * 1.5 = 10.5 dollars
the cost to maintain the tree per year is 3 dollars
so the net income per year is 10.5 - 3 = 7.5 dollars
to find out how many years it will take for Carlos to start earning money, we need to find the point where the total income from planting the tree is greater than the initial cost
let x be the number of years it takes to start earning money
the total income after x years is 7.5x
the total cost after x years is 90 + 3x
we want to find the point where 7.5x > 90 + 3x
subtracting 3x from both sides gives us 4.5x > 90
dividing both sides by 4.5 gives us x > 20
since x represents the number of years, it must be a whole number
so it will take at least 21 years for Carlos to start earning money on the lemon tree
</think><answer>
21
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